Aim: Study of Electromatic spectrum Allocation & wireless Bands used in communication

Theory:

1. The Physics: Core Electromagnetic Theory

Wireless communication relies on **electromagnetic waves** traveling through space. Two fundamental properties dictate how a wave behaves:

- **Frequency (f):** The number of wave cycles per second, measured in Hertz (Hz).
- **Wavelength (λ):** The physical distance between two wave peaks.

Because the speed of light (c) is constant, frequency and wavelength are inversely proportional ($\lambda = c / f$). This relationship creates a critical physical trade-off:

- **Low Frequencies (Long Wavelengths):** Can travel long distances and easily pass through solid obstacles like walls, buildings, and trees. However, they have less capacity to carry large amounts of data.
- **High Frequencies (Short Wavelengths):** Can carry massive amounts of data at ultra-fast speeds. However, they travel very short distances and are easily blocked or absorbed by rain, foliage, and concrete walls.

2. Wireless Bands Theory: The Spectrum Segments

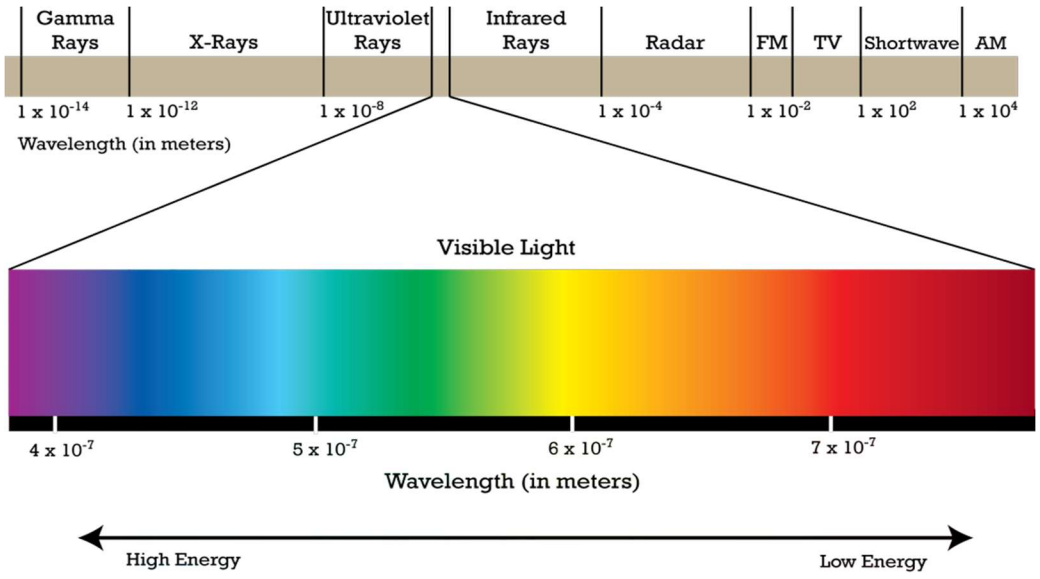
The radio spectrum (roughly 3 kHz to 300 GHz) is divided into distinct "bands." Each band has unique characteristics that make it ideal for specific wireless technologies:

Band Range	Propagation Traits	Primary Applications
Low-Band *(Below 1 GHz)*	High range, excellent indoor penetration, low data capacity.	AM/FM radio, television broadcasting, early LTE (4G), IoT devices.
Mid-Band *(1 GHz – 6 GHz)*	Balanced range and capacity. The "sweet spot" for modern data.	**Wi-Fi** (2.4 GHz & 5 GHz), **Bluetooth**, **5G sub-6**, GPS, radar.
High-Band / mmWave *(24 GHz – 100 GHz)*	Ultra-high speeds, near-zero lag, incredibly poor range/penetration.	Dense urban 5G nodes, fixed wireless internet, airport security scanners.

3. Key Concepts in Allocation Theory

Because two transmitters using the exact same frequency in the same area will cancel each other out (interference), regulatory bodies use specific allocation strategies:

- **Channel Bandwidth:** Think of a frequency band as a highway. The bandwidth is the number of lanes. A wider channel (e.g., 80 MHz vs. 20 MHz) allows more data to flow simultaneously, which is why 5G and modern Wi-Fi use massive channel widths to achieve gigabit speeds.
- **Duplexing (Frequency vs. Time):** To allow two-way communication (downloading and uploading), systems split the spectrum:
    - o **FDD (Frequency Division Duplex):** Uses two separate frequencies (one for talking, one for listening).
    - o **TDD (Time Division Duplex):** Uses a single frequency but alternates between talking and listening in microsecond intervals.
- **Licensed vs. Unlicensed Spectrum:**
    - o **Licensed:** Operators (like Airtel, Jio, or Verizon) pay billions in government auctions for exclusive rights to a frequency block. This guarantees zero interference for their users.
    - o **Unlicensed (ISM Bands):** The Industrial, Scientific, and Medical bands are free for anyone to use. Wi-Fi, Bluetooth, and microwaves share this space. To prevent total gridlock, devices must adhere to strict transmission power limits.



EM Spectrum Region	Frequency Range	Wavelength Range	Core Characteristics & Wave Behavior	Common Applications
Radio Waves	< 3 GHz	> 10 cm	Longest wavelengths; easily passes through walls and atmosphere; lowest energy.	Cellular networks (4G/5G), Wi-Fi, AM/FM radio, GPS navigation.
Microwaves	3 GHz – 300 GHz	1 mm – 10 cm	High-frequency radio subset; absorbed by water molecules; requires line-of-sight.	Microwave ovens, satellite communications, radar tracking systems.
Infrared (IR)	300 GHz – 400 THz	750 nm – 1 mm	Experienced primarily as heat; blocked by solid objects; invisible to human eyes.	TV remote controls, night-vision goggles, thermal imaging, fiber-optic internet.
Visible Light	400 THz – 790 THz	380 nm – 750 nm	The only sliver of the spectrum visible to humans; ranges from Red (lowest energy) to Violet (highest).	Human vision, photography, LED lighting, laser pointers.
Ultraviolet (UV)	790 THz – 30 PHz	10 nm – 380 nm	High enough energy to cause chemical reactions and sunburns; mostly absorbed by Earth's ozone layer.	Sterilization/sanitizing lamps, counterfeit currency detection, black lights.
X-Rays	30 PHz – 30 EHz	0.01 nm – 10 nm	High-energy ionizing radiation; easily penetrates soft tissues but is stopped by dense materials like bone.	Medical imaging (radiography), airport luggage scanners, structural integrity testing.
Gamma Rays	> 30 EHz	< 0.01 nm	Shortest wavelengths and highest energy; highly penetrating and dangerous ionizing radiation.	Cancer radiation therapy, nuclear medicine, astronomical star observation.